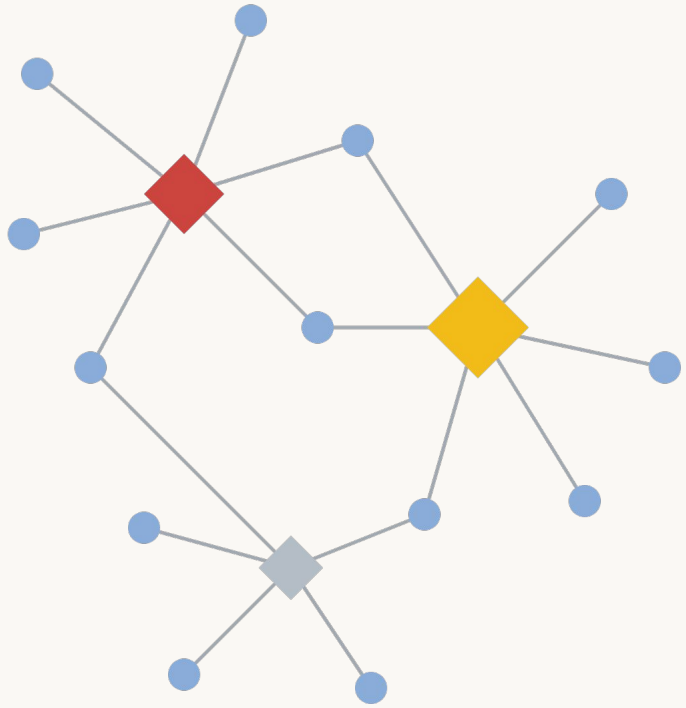
Genes do not act alone



Module 3 · Gene regulatory networks, their properties and how to infer them from expression data

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AORN Santobono-Pausilipon IRCCS · BBCC2026 tutorial · TARGET-WT, 82 favorable, 42 unfavorable, 6 normal kidneys

From one gene at a time to genes as a system

Module 2, limma

Is this gene different from normal kidney?

Each gene is tested alone. The result is a ranked list that says nothing about whether two candidates influence each other.

Module 3, GENIE3

How do they work together, and does the wiring change?

The same genes are read as a network, in three conditions.

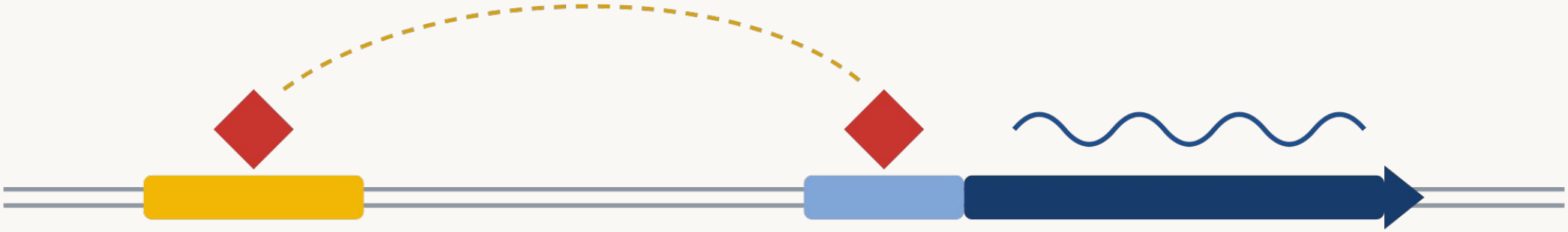
three networks on the same genes → compare edges and hubs

What a gene regulatory network is

The set of interactions between transcription factors, cis-regulatory DNA and target genes that decides which genes are on, how much, where and when. It shapes cell identity and fate, and it is rewired in disease.

as biology	proteins binding DNA, activating or repressing transcription
as mathematics	a directed, weighted graph, with genes as nodes and regulations as edges
as inferred here	from RNA-seq alone, so every edge is a statistical hypothesis, not a measured binding event

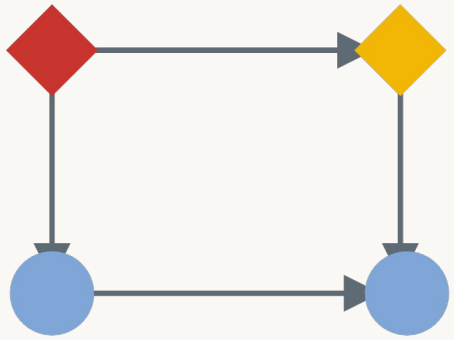
Three players behind every edge



Red diamonds, TFs. Yellow, an enhancer. Light blue, the promoter. Dark blue, the target gene and, wavy, its mRNA. Dashed, the enhancer–promoter loop.

Transcription factor	binds a DNA motif; there are 1,639 human TFs (Lambert et al., 2018)
Cis-regulatory element	a promoter or an enhancer, bound only where chromatin is accessible
Target gene	its transcription goes up with activation and down with repression

A network is a list of arrows



A top left, B top right, C bottom left, D bottom right. Diamonds are TFs.

Genes are the nodes.

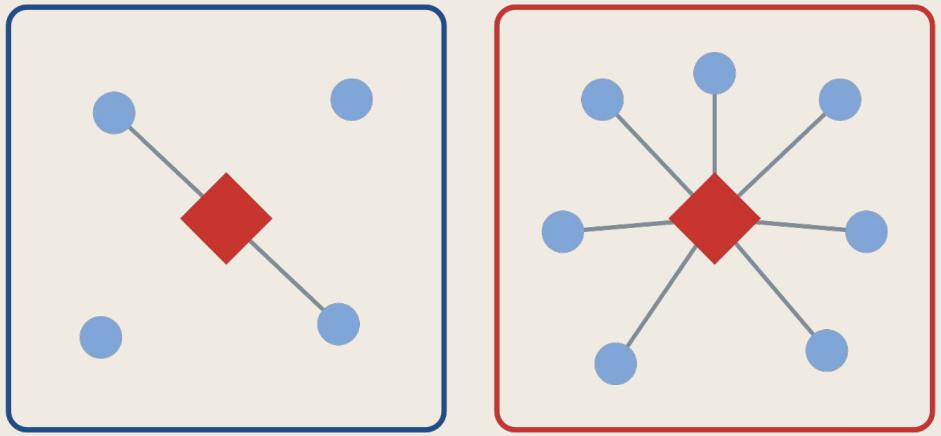
An arrow from A to B means “A regulates B”, so direction matters.

	A	B	C	D
A	0	1	1	0
B	0	0	0	1
C	0	0	0	1
D	0	0	0	0

The same network as a table.

1 means the gene in the row regulates the gene in the column.

The same genes, wired differently



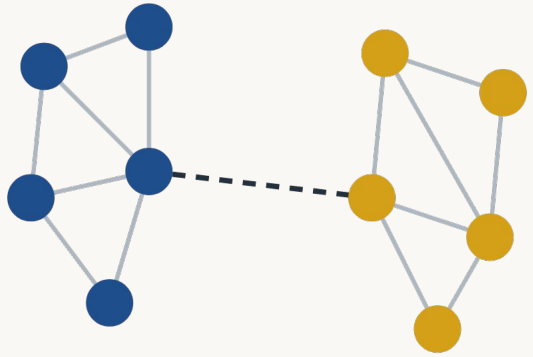
Same TF, same genes. Condition A on the left with 2 targets, condition B on the right with 7.

A TF acts only where its partners, the chromatin of its targets and its own activity allow. A network belongs to a cell type and a condition.

Edges can be gained or lost between conditions even when the TF's own expression does not change.

Module 2's per-gene test cannot see this, which is why Module 3 builds one network per condition.

Genes work in programmes



Two modules, dense inside, joined by one dashed edge.

A module is a group of genes with many links inside and few links outside.

Modules map to functions, such as the cell cycle, the immune response or nephron development.

They are often layered, with master TFs regulating other TFs that regulate the working genes.

The E2F1 targets at the end of this talk look exactly like one module, the replication and mitosis programme.

How to infer a network

expression data → a score per pair → a network

genes measured in many samples

how much one gene tells us about another

keep only the strongest links

The methods differ only in the middle step, in how they compute that score.

Many more unknowns than samples

every TF could regulate every gene, so with all human TFs and genes there are about

30 million

possible arrows

Reduce the space

Module 3 keeps strong DEGs only, |log2FC| > 3, and lets only TFs be regulators.

Keep only the strongest

Real networks are sparse, so we keep the top 300 links.

Accept uncertainty

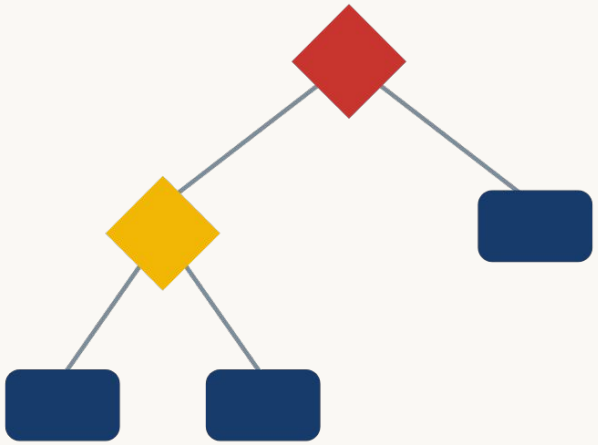
Edges are ranked hypotheses. Benchmarks show low accuracy, worse with many genes.

Five ways to score a link

family	the score asks	examples	watch out for
correlation	do the two genes go up and down together?	WGCNA, ARACNE	no direction; indirect links
partial correlation	do they still, once all other genes are accounted for?	graphical lasso	needs many samples
regression	how well does TF i predict gene j?	GENIE3, LASSO, SCENIC	similar TFs share the credit
probabilistic	how likely is this arrow, given the data?	Bayesian networks	no feedback loops; slow
dynamical	does TF i change the rate at which j is made?	ODE models	needs time courses

Categories after Kim et al., npj Syst Biol Appl 2023

Predicting each gene from the TFs



For every gene, f is a regression model that try to guess the gene's expression from the expression of all the TFs.

A TF that helps the guess a lot gets a lot of credit.
That credit is the strength of the arrow from the TF to the gene.

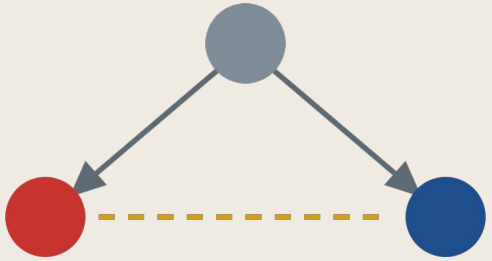
A direct regulator usually predicts its target better than an indirect one, which is why prediction beats simple correlation.

Co-expression is not regulation



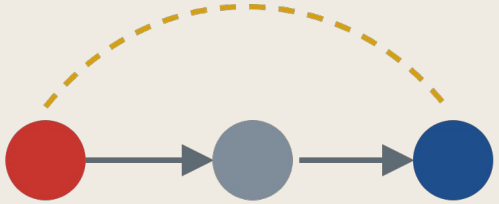
Which way?

Two correlated TFs, and the data alone cannot say which one regulates the other.



A hidden regulator

One factor drives both genes, which correlate without ever touching.



Direct or indirect?

In a chain the two ends correlate. Conditioning helps, as in partial correlation and regression.

In Module 3 the direction is imposed, since only TFs may regulate.

Causal evidence needs chromatin data or perturbation.